

# Lepton Empirics Check: the FFGFT Relations Seen from the Data

Koide ratio, circulant angle and  $\xi$  ladder computed backwards from the PDG masses – with the pre-booked prediction  $m_\tau = 1776.968$  MeV

Johann Pascher  
ORCID: 0009-0000-6518-4064

2 July 2026 | Doc. 292

## Contents

|    |                                                                                                           |    |
|----|-----------------------------------------------------------------------------------------------------------|----|
| .1 | Origin: how $2/9$ became visible in the first place . . . . .                                             | 2  |
| .2 | Purpose and direction of the calculation . . . . .                                                        | 2  |
| .3 | Level 1: the circulant – Koide relation and angle . . . . .                                               | 2  |
|    | Koide ratio from the data . . . . .                                                                       | 2  |
|    | Circulant fit: amplitude and angle from three masses . . . . .                                            | 3  |
|    | The $\tau$ -free test: angle from $m_\mu/m_e$ alone . . . . .                                             | 3  |
|    | Forward check and the one prediction . . . . .                                                            | 3  |
| .4 | Level 2: the $\xi$ ladder – ratios and reverse $\xi$ . . . . .                                            | 4  |
|    | Ladder ratios against the data . . . . .                                                                  | 4  |
|    | Reverse $\xi$ : what the data return for $\xi$ . . . . .                                                  | 4  |
|    | Bridge constants per P40 . . . . .                                                                        | 4  |
|    | $K_{\text{frak}}$ cross-check: no simple pattern . . . . .                                                | 4  |
|    | Cross-ladder coupling: a generation-linear correction law . . . . .                                       | 5  |
| .5 | Relation to the measurement precisions . . . . .                                                          | 6  |
|    | Thought model: inverting the precision statement . . . . .                                                | 7  |
|    | Resolved by declaration: the reference point $m_\mu/m_e$ (P42) . . . . .                                  | 7  |
| .6 | The fine-structure constant from the lepton masses . . . . .                                              | 8  |
|    | The $\alpha$ residual and the dressing layer . . . . .                                                    | 8  |
| .7 | Two layers that do not need each other: what $2/9$ provides and what the SI conversion provides . . . . . | 9  |
| .8 | Placement . . . . .                                                                                       | 10 |
|    | The residual as the signature of incompleteness . . . . .                                                 | 10 |

## .1 Origin: how $2/9$ became visible in the first place

Before any calculation, the question of where the central number of this document comes from should be settled – because it was not sought but found. The route was the Hilbert-space translation of FFGFT. The bridge of Docs. 230/231/232 maps the loop observables of the corpus to self-adjoint operators on a fibre-valued Hilbert space; only when this translation was carried through to the masses (Doc. 282) did the decisive structure appear: the three lepton masses are the spectrum of a hermitian  $Z_3$  circulant on the  $\mathbb{C}^3$  fibre of the full space  $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$ . In the eigenvalues of this operator the phase angle  $\theta = 2/9$  appears – not as an ansatz one inserts, but as a value the diagonalisation returns.  $2/9$  is therefore *not* a starting point of the mass calculation but an order that became visible only afterwards: it was the change of representation – from the  $\xi$  ladder to the operator/spectral view – that exposed it. This counter-calculation therefore tests not an input but a *discovery*; that is why the direction from the data (below) is meaningful at all. The relation to the  $\xi$  ladder, which computes the masses without  $2/9$ , is clarified in the closing section “Two layers”.

## .2 Purpose and direction of the calculation

The FFGFT corpus fundamentally computes *away from*  $\xi = 4/30000$ : the geometry yields relations that are subsequently checked against data. This document reverses the direction. The starting point is exclusively the empirical lepton masses (PDG), and the question is what the FFGFT relations of the lepton sector make *of the data* – which structural numbers the data return, with which uncertainties, and where residual tensions sit. The mode is the compatibility mode: no new derivation, but an independent counter-calculation per P35, with the level separation per P40 (only ratios are exact; absolute values carry the bridge constants). All numbers in this document come from `python/Dok292_Skripte/leptonen_empirie_check.py` (numpy-only, seed 20780458) and are reproducible with a single call.

The inputs are

$$m_e = 0.51099895069(16) \text{ MeV}, \quad (1)$$

$$m_\mu = 105.6583755(23) \text{ MeV}, \quad (2)$$

$$m_\tau = 1776.86(12) \text{ MeV}. \quad (3)$$

The value for  $m_\tau$  is the PDG 2022 value; the numbers in the text are computed with it. PDG 2024 gives  $m_\tau = 1776.93(9) \text{ MeV}$ . This yields  $Q = 0.6666645$ ,  $Q - 2/3 = -2.2 \cdot 10^{-6} = -0.43 \sigma$ , and  $m_\tau^{\text{pred}} = 1776.968 \text{ MeV}$  lies  $+0.43 \sigma$  above the measured value. The “single  $\tau$  tension of  $\sim 0.9 \sigma$ ” referred to below and the  $\sigma$  figures derived from it hold for the PDG 2022 value and roughly halve with PDG 2024. The  $\tau$  mass is by orders of magnitude the least precise of the three; as will become clear, it dominates all residual tensions.

## .3 Level 1: the circulant – Koide relation and angle

### Koide ratio from the data

The Koide ratio  $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$  evaluated on the PDG values gives

$$Q = 0.666660511 \pm 0.0000068, \quad (4)$$

where the uncertainty is carried almost entirely by  $m_\tau$ . The deviation from the exact value  $2/3$  is  $-6.2 \cdot 10^{-6}$ , i.e.  $-0.91 \sigma$  (with PDG 2024:  $-2.2 \cdot 10^{-6}$ ,  $-0.43 \sigma$ ). The data are compatible

with exactly  $2/3$ ; they do not *enforce* the exact value, but they do not contradict it at 5–6 significant digits.

### Circulant fit: amplitude and angle from three masses

Following Doc. 282, the  $\sqrt{m_k}$  are eigenvalues of a hermitian  $Z_3$  circulant,  $\sqrt{m_k} = M(1 + r\cos(\theta + 2\pi k/3))$ , with the two pure structural numbers  $r = \sqrt{2}$  and  $\theta = 2/9$  and the identity  $Q = (1 + r^2/2)/3$ . The Fourier extraction from the three data points yields (assignment  $k = 0, 1, 2 \rightarrow \tau, e, \mu$ ; all cyclic assignments give the same angle)

$$M = 313.84 \text{ MeV}, \quad r_{\text{emp}} = 1.4142005, \quad \theta_{\text{emp}} = 0.22222963 \pm 8.4 \cdot 10^{-6}. \quad (5)$$

The amplitude deviates from  $\sqrt{2} = 1.4142136$  by  $1.3 \cdot 10^{-5}$  – via  $Q = (1 + r^2/2)/3$  this is the *same* statement as the  $Q$  deviation, not an independent one. The angle lies  $+0.89\sigma$  above  $2/9 = 0.22222222$ , likewise  $\tau$ -limited.

### The $\tau$ -free test: angle from $m_\mu/m_e$ alone

Doc. 282 names  $m_\mu/m_e$  as the sharpest test of whether  $\theta$  is exactly  $2/9$ : holding  $r = \sqrt{2}$  fixed,  $M$  cancels in the ratio, and the two most precise masses in physics determine  $\theta$  alone. Solving  $(1 + \sqrt{2}\cos(\theta + 4\pi/3))^2 / (1 + \sqrt{2}\cos(\theta + 2\pi/3))^2 = m_\mu/m_e$  gives

$$\theta(\mu/e) = 0.2222220471, \quad |\theta - 2/9| = 1.75 \cdot 10^{-7}. \quad (6)$$

This independently reproduces the figure from Doc. 282 ( $\approx 1.8 \cdot 10^{-7}$ , seven significant digits). Two readings must be kept apart. Measured against the tiny PDG uncertainty of  $m_\mu/m_e$ , the residual is formally highly significant – *under the additional assumption* that  $r$  is exactly  $\sqrt{2}$ . If  $r$  is allowed to float via  $Q$ , the residual is not an independent tension but once again the same single  $\tau$  tension of  $\sim 0.9\sigma$ . In the lepton sector there are thus not four small deviations but *one*, appearing in four guises.

### Forward check and the one prediction

Conversely, setting  $r = \sqrt{2}$  and  $\theta = 2/9$  exactly, the circulant reproduces  $m_\mu/m_e$  to  $+0.0010\%$  and  $m_\tau/m_\mu$  to  $+0.0060\%$ . Inverting for  $m_\tau$  turns this into a falsifiable forward statement: from  $(m_e, m_\mu)$  and the two structural numbers it follows that

$$m_\tau^{\text{pred}} = 1776.968 \text{ MeV} \quad (7)$$

(the three inversion routes –  $Q = 2/3$  exact,  $\theta = 2/9$  exact in the three-mass fit, direct prediction from  $(m_e, m_\mu; \sqrt{2}, 2/9)$  – give 1776.969, 1776.967 and 1776.968 MeV, i.e. the same value). Against PDG 2022  $1776.86 \pm 0.12$  MeV this is  $+0.90\sigma$ , against PDG 2024  $1776.93 \pm 0.09$  MeV  $+0.43\sigma$ . A  $\tau$ -mass measurement at the targeted Belle-II level of  $\sim 0.05$  MeV decides this tension: if the value lands near  $\approx 1776.97$ , the  $\tau$  direction is resolved and the pair  $(\sqrt{2}, 2/9)$  is confirmed at the  $10^{-5}$  level; if it lands significantly below, the approximation at this level is refuted. This is a genuine prediction in the sense of the corpus methodology, not a retrodiction: the value is fixed before the measurement and is booked here in advance. What the test does *not* decide is shown by the precision analysis in Section .5.

## .4 Level 2: the $\xi$ ladder – ratios and reverse $\xi$

### Ladder ratios against the data

The ladder representation  $m_i = r_i \xi^{p_i} v$  (Docs. 006/046; leptons:  $r_e = 4/3$ ,  $p_e = 3/2$ ;  $r_\mu = 16/5$ ,  $p_\mu = 1$ ;  $r_\tau = 25/9$ ,  $p_\tau = 2/3$  per K2) makes the ratios parameter-free:

$$\left. \frac{m_\mu}{m_e} \right|_{\text{ladder}} = \frac{12}{5} \xi^{-1/2} = 207.85 \quad (\text{PDG } 206.77; +0.52 \%), \quad (8)$$

$$\left. \frac{m_\tau}{m_\mu} \right|_{\text{ladder}} = \frac{125}{144} \xi^{-1/3} = 16.992 \quad (\text{PDG } 16.817; +1.04 \%), \quad (9)$$

$$\left. \frac{m_\tau}{m_e} \right|_{\text{ladder}} = \frac{25}{12} \xi^{-5/6} = 3532 \quad (\text{PDG } 3477; +1.57 \%). \quad (10)$$

The ladder thus carries the orders of magnitude and the  $\sim 1\%$  accuracy – not the precision of the circulant. Both levels are to be read separately, as declared in Doc. 282 and P40: the circulant delivers four to five digits from  $(\sqrt{2}, 2/9)$ , the ladder delivers the hierarchy from  $\xi$ .

### Reverse $\xi$ : what the data return for $\xi$

Solving the three ladder ratios for  $\xi$ , the data return

$$\xi_{\mu/e} = 1.347 \cdot 10^{-4}, \quad \xi_{\tau/\mu} = 1.375 \cdot 10^{-4}, \quad \xi_{\tau/e} = 1.358 \cdot 10^{-4}, \quad (11)$$

i.e.  $+1.1$  to  $+3.2\%$  above  $4/30000 = 1.3333 \cdot 10^{-4}$ . The three values scatter among each other by  $\sim 2\%$  – the ladder is consistent in the ratios at the percent level, not exact. This is the honest empirical status of the ladder representation, unchanged from the declaration in Doc. 006 (mean deviation below  $1\%$  at the mass level; correspondingly  $1\text{--}3\%$  at the ratio level).

### Bridge constants per P40

Per lepton, solving for the bridge constant  $v_{\text{eff}} = m_i / (r_i \xi^{p_i})$  yields

$$v_{\text{eff}}^{(e)} = 248.9 \text{ GeV}, \quad v_{\text{eff}}^{(\mu)} = 247.6 \text{ GeV}, \quad v_{\text{eff}}^{(\tau)} = 245.1 \text{ GeV}, \quad (12)$$

scattered around the Higgs vacuum expectation value  $246.22 \text{ GeV}$  ( $+1.10\%$  /  $+0.58\%$  /  $-0.46\%$ ). Equivalently: the required  $r_{\text{eff}}$  at fixed  $v$  deviate by the same percentages from the declared rational values ( $r_e$ :  $1.348$  instead of  $4/3$ , etc.). This is precisely the P40 finding seen from the data side: the rational structural numbers of the ladder hold at the percent level; the residual deviation is absorbed into the bridge constant and is *not* fully cancelled in ratios.

### $K_{\text{frak}}$ cross-check: no simple pattern

The obvious question of whether the ladder residuals follow the fractal correction factor is answered in the negative by the calculation:  $K_{\text{frak}} = 1 - 100\xi$  corresponds to  $-1.333\%$ , the half and three-halves powers to  $-0.665$  and  $-1.995\%$  – the measured deviations  $+0.52$  and  $+1.04\%$  fit no single  $K_{\text{frak}}$  power in either magnitude or sign. The only striking feature is that the  $\tau/\mu$  deviation is quite precisely twice the  $\mu/e$  deviation; whether structure or coincidence lies behind this remains unbooked per P35. The point is declared open, not explained away.

## Cross-ladder coupling: a generation-linear correction law

Before this finding is booked, the booking level must be fixed, since it decides admissibility. There are two *mutually exclusive* calculation modes. *Mode 1* computes all lepton ratios purely from  $\xi$ , without a reference point: then  $\mu/e$ ,  $\tau/\mu$  and  $\tau/e$  are all genuine predictions (each 0.5–1.6 % off the data), and all three are legitimate test points. *Mode 2* applies P42:  $m_\mu/m_e$  is then the declared anchor, *consumed* and no longer a test quantity – one may no longer book its ladder deviation, else the same quantity is used twice (once as anchor, once as test). The coupling analysis below belongs *exclusively* to Mode 1, the reference-free one; in Mode 2 it is void, because there  $\mu/e$  drops out of the comparison and the only independent ladder test left is  $m_\tau$  – which in any case runs through the circulant, not the coarse ladder. Within Mode 1: the statement “no simple pattern” concerns a *single*  $K_{\text{frak}}$  power. Asking instead whether the ladder stages are coupled to *each other* does reveal structure. Writing the required correction per place as an effective factor  $N_g$  via  $K = 1 - N_g \xi$  (or as an exponent  $p$  via  $K_{\text{frak}}^p$ ), the data give

$$\mu/e: \quad N_1 = 38.89, \quad p_1 = 0.3873, \quad (13)$$

$$\tau/\mu: \quad N_2 = 77.06, \quad p_2 = 0.7694, \quad (14)$$

$$\tau/e: \quad N_3 = 115.55, \quad p_3 = 1.1567. \quad (15)$$

Two relations here are not accidental. First, the exponent structure is *multiplicatively consistent*:  $p_3 = p_1 + p_2$  to  $6 \cdot 10^{-14}$ . The  $\tau/e$  place therefore carries *no* new information – it is the product of the two stages, as a throughgoing multiplicative correction  $K^{p_1} \cdot K^{p_2} = K^{p_1+p_2}$  requires. Second, the effective factors stand in the ratio  $N_1 : N_2 : N_3 = 1 : 1.981 : 2.971 \approx 1 : 2 : 3$ : the correction grows *linearly with generation number*,  $N_g \approx g \cdot N_0$  with  $N_0 \approx 38.6$  (the three individual estimates  $N_1$ ,  $N_2/2$ ,  $N_3/3$  are 38.89, 38.53, 38.52 – consistent among themselves to below 1 %).

This is the structural reason why a constant  $K_{\text{frak}}$  does not close the ladder, and why asking for the “right” number instead of 100 (such as 89) misses the point: the ladder demands not a fixed factor but a *generation-counting* one,  $N_g = g \cdot N_0$ . At  $\alpha$  only one scale is in play (the  $\mu e$  mean) – one factor suffices, and  $\alpha$  fits. In the ladder three generations superpose, so  $N$  grows linearly; a constant factor cannot do this in principle. The generation-dependent dressing named earlier as an obligation thereby acquires a concrete form: to first approximation it is linear in  $g$ .

The tolerance balance shows how far the law reaches (reference-free Mode 1, common  $N_0 = 38.65$ ):

| Place      | raw      | raw [ $\sigma_{\text{exp}}$ ] | corrected | corr. [ $\sigma_{\text{exp}}$ ] |
|------------|----------|-------------------------------|-----------|---------------------------------|
| $\mu/e$    | +0.521 % | $2.4 \cdot 10^5$              | +0.0033 % | $1.5 \cdot 10^3$                |
| $\tau/\mu$ | +1.038 % | 154                           | −0.0031 % | 0.47                            |
| $\tau/e$   | +1.565 % | 232                           | −0.0052 % | 0.78                            |

The law reduces the ladder deviation by about a factor of 300, from 0.5–1.6 % to  $\sim 5 \cdot 10^{-3}$  %. The two  $\tau$  places thereby fall *within* their measurement tolerance ( $< 1 \sigma$ ) – for them the law is practically complete. The  $\mu/e$  entry stays at  $\sim 1.5 \cdot 10^3 \sigma$ , not because the correction fails there but because  $\mu/e$  is measured about 500 times more precisely: even a 0.003 % residual is enormous in  $\sigma$  units. This is exactly where the mode boundary bites: this  $\mu/e$  entry is a statement *only* in reference-free Mode 1; under P42,  $\mu/e$  is the anchor and drops out of the balance. The remaining residual of the two  $\tau$  places sits in the fine deviation of the ratio from exact  $1 : 2 : 3$  ( $\sim 1$ –2 %) and in the still-undetermined base unit  $N_0$  – that is the next open calculation, not a contradiction.

Open – and per P35 expressly not explained away – are two things. The base unit  $N_0 \approx 38.6$  is not yet derived from the geometry; obvious candidates ( $100/e = 36.8$ ,  $100/\varphi^2 = 38.2$ , plain

39) do not match it closely enough to be claimed. And the phase sector – the  $\delta$  fine value from Doc. 291 with  $p_\delta = -0.88$  – does *not* obviously fit the same generation-linear magnitude ladder; whether the magnitude and phase sectors couple through a common law is a separate question not answered here. The finding is thus not the solution of the ladder residual but its sharpening: “three crooked exponents” becomes “one generation-linear law  $N_g = g \cdot N_0$  with  $N_0$  still undetermined”.

The limit of this claim, following from the mode separation, must be stated explicitly. The 1 : 2 : 3 pattern is supported at three points *only* in the reference-free Mode 1. As soon as P42 is applied and  $m_\mu/m_e$  is declared the anchor (the level needed for the circulant precision and the  $m_\tau$  prediction),  $\mu/e$  drops out as a test point, and the law rests only on  $\tau/\mu$  and  $\tau/e$  – which share the same  $\tau$  mass and whose precision is low. The apparently threefold confirmation is thus tied to reference-freeness. Both at once –  $\mu/e$  as anchor *and* as an  $N_g$  test point – is inadmissible double-booking. The circulant, which already carries  $\mu/e$  to  $10^{-5}$  (route A in Section .5), does not need the  $g$  correction anyway; it is a statement about the coarse  $\xi$  ladder in the reference-free mode alone, not about the precise circulant layer.

A common misunderstanding must be cleared up here: the 100 in  $K_{\text{frak}} = 1 - 100\xi$  and the base unit  $N_0 \approx 38.6$  are *not* the same quantity, so the ladder by no means needs “fewer recursions” than the  $\alpha$  sector. The 100 is the single fixed scale of a corpus-wide constant (X-flip,  $T_{\text{CMB}}$ ,  $E_0$  bridge);  $N_0$  is the factor *per generation* in a different sector. The difference is not *fewer* but *other*: not a fixed factor but a generation-counting  $N_g = g \cdot N_0$  – already 77 at  $g = 2$ , 116 at  $g = 3$ , i.e. partly *more* than 100. Comparing the two numbers as “recursion depths” therefore misses the point.

In assessing possible expressions for  $N_0$  the correct measure is not point-exact agreement – that is unreachable in principle – but consistency *within the tolerance* to which  $N_0$  can be determined at all. That tolerance is considerable: in the referenced mode  $N_0$  follows from the  $\tau$  places alone, and their precision is limited by the  $\tau$  mass ( $\pm 0.12$  MeV). Propagating this gives  $N_0 = 38.52 \pm 0.21$ , a relative determination uncertainty of about 0.5 %. A candidate expression is therefore to be judged not by whether it hits 38.6 to the decimal, but by whether it falls into this interval.

By that measure – and still as an *unproven* observation per P35, not a derivation –  $100/\varphi^2 = 38.20$  is an *admissible* candidate: its distance to  $N_0$  is  $1.6 \sigma_{N_0}$ , i.e. within the tolerance. As a point value the expression misses 38.6; within the tolerance to which  $N_0$  is known, it matches. Were  $N_0 = 100/\varphi^2$ , the ladder base unit would follow from the  $K_{\text{frak}}$  scale through a single factor  $\varphi^{-2}$  – magnitude and correction sectors hanging on the same constant. The finding is thus not “narrowly missed and discarded” but “compatible with the present determination, not refuted”. A sharper  $\tau$  mass (Belle II) or a geometric derivation of  $N_0$  will decide between  $100/\varphi^2$  and the alternatives; until then it remains an admissible, not a proven, expression.

## .5 Relation to the measurement precisions

The  $\sigma$  statements so far refer to one quantity each. The full question is: how do the FFGFT structural numbers sit relative to the *entire* precision structure of the data? A two-parameter analysis answers it. The two independent ratios  $R_1 = m_\mu/m_e$  and  $R_2 = m_\tau/m_\mu$  determine the pair  $(r, \theta)$  exactly; their relative measurement precisions, however, differ dramatically:  $\sigma_{R_1}/R_1 = 2.2 \cdot 10^{-8}$  versus  $\sigma_{R_2}/R_2 = 6.8 \cdot 10^{-5}$  – the  $\mu/e$  ratio is measured about 3100 times more precisely. Error propagation onto  $(r, \theta)$  therefore yields an extremely anisotropic covariance: a *thin* principal direction with  $\sigma \approx 3.4 \cdot 10^{-10}$  (carried by  $\mu/e$ ) and a *thick* one with  $\sigma \approx 1.7 \cdot 10^{-5}$  (carried by  $\tau$ ); the correlation between  $r$  and  $\theta$  is practically  $-1$ .

In this picture the point  $(\sqrt{2}, 2/9)$  lies away from the data fit  $(1.41420051, 0.22222963)$  by  $0.90\sigma$  in the thick direction – this is the single  $\tau$  tension of Section 2 – and by about  $450\sigma$  in the thin one. Spelled out: the circulant with *exactly*  $(\sqrt{2}, 2/9)$  misses  $m_\mu/m_e$  by  $1.0 \cdot 10^{-5}$  relative, while the measurement stands at  $2.2 \cdot 10^{-8}$ . As an exact pair the two structural numbers are thus already excluded by today's data; they are confirmed at the  $10^{-5}$  level – precisely the “four to five digits” declared in Doc. 282, here made precise as a measurement relation. This also bounds the reach of the  $m_\tau$  prediction: a perfect Belle-II result resolves only the thick  $\tau$  direction; the residual of  $1.75 \cdot 10^{-7}$  in  $\theta$  (equivalently  $10^{-5}$  in  $R_1$ ) is an *already measured*, standing offset that any exactness claim would have to explain – radiative corrections to the pole masses are the obvious candidate, though their typical size ( $\alpha/\pi \sim 10^{-3}$ ) rather deepens the puzzle of why the agreement reaches  $10^{-5}$  at all. The point is booked open per P35.

### Thought model: inverting the precision statement

The  $450\sigma$  finding rests entirely on trust in the quoted precision of  $m_\mu/m_e$ . The inversion – the precision might not mean what it claims – is admissible as a thought model and comes in three strengths. *Weak*: the experimental uncertainty would be underestimated by a factor of  $\sim 450$  – quantitatively hardly tenable, since the same precision hangs in a web of independent cross-checks (muonium spectroscopy,  $g-2$ , QED tests). *Intermediate*:  $m_\mu/m_e$  is not a read-off but a quantity *extracted* through bound-state QED; the quoted precision presupposes the completeness of the extraction theory, and a missing term at  $10^{-5}$  would dissolve the conflict – 450 times the estimated theory error, borderline, but logically open. *Strong (and FFGFT-native)*: the precision is real but applies to the *dressed* quantity – the pole masses in the SI frame – while the circulant relation would live on the geometric level beneath. Then nothing contradicts anything: the  $10^{-5}$  offset would not be an error but the *measured thickness of the dressing layer* in the  $\mu/e$  channel. This is the consistent extension of P40: in ratios only the generation-*independent* part of the fractal/SI correction cancels; a generation-dependent remainder survives – and would have exactly the shape of this offset. The price of the strong reading is an obligation: the offset would have to be computable from the dressing (generation-dependent radiative structure), otherwise the reading is unfalsifiable. It is booked here as a thought model, not as a resolution; the contradiction finding of the previous section stands for the pole-mass level.

### Resolved by declaration: the reference point $m_\mu/m_e$ (P42)

The framework decides the question not through one of the thought models but through the claim layer: FFGFT fundamentally keeps the purely theoretical derivation from  $\xi$ , and the lepton sector receives – analogous to the cosmic scale (R39) – exactly *one* declared reference point, the ratio  $m_\mu/m_e$ . At structural  $r = \sqrt{2}$  (from  $Q = 2/3$ ) it fixes the operative angle  $\theta_{\text{ref}} = 0.2222220471$ ; the structural number  $2/9$  is its rational address to seven digits, not an exactness claim at the pole-mass level. The  $450\sigma$  finding is thereby no longer a falsification but part of the calibration. Two delimitations: the status of  $\xi$  is untouched (FFGFT claims  $\xi$  as forced by the geometry; that dispute is not adjudicated here), and the lepton masses are in themselves not an input of the theory – the reference point belongs to the comparison level, it locates the geometric relation in the pole-mass data and would not be needed without the intent to compare. The honest price lies in the test balance:  $m_\mu/m_e$  is consumed as anchor and drops out as a test quantity; the test quantity is  $m_\tau$  alone. The gain is an input-error-free sharp prediction:

$$m_\tau = 1776.9690 \pm 0.0001 \text{ MeV}, \quad (16)$$

where the uncertainty stems solely from  $(m_e, m_\mu)$  and practically vanishes; against PDG the single  $\tau$  tension remains  $(+0.9\sigma$  with PDG 2022,  $+0.43\sigma$  with PDG 2024), and the Belle-II world average decides with no escape route. For the bridge question (Doc. 291) the declaration sharpens the target: the dynamics must select  $\theta_{\text{ref}}$ ;  $2/9$  remains its address and the pre-registered Stage-10 target, whose orbit resolution ( $\sim 0.1$ ) is unaffected by the  $10^{-7}$  difference. Booked as P42 in Doc. 190.

The same view calibrates the  $\xi$  ladder: its residuals correspond to  $2.4 \cdot 10^5$  ( $\mu/e$ ), 154 ( $\tau/\mu$ ) and 232 ( $\tau/e$ ) experimental standard deviations. The ladder residuals are therefore not data-limited but theory-side (P40 bridge constants); measurement precision has outrun the ladder representation by two to five orders of magnitude. Finally, the test power of the  $m_\tau$  prediction (separation of prediction from today's central value) at  $\sigma_{m_\tau} = 0.12/0.05/0.02$  MeV is  $0.9/2.2/5.4\sigma$  respectively (relative to the PDG 2022 central value; roughly halved with PDG 2024) – a single Belle-II measurement is indicative; only the combined world average is decisive.

## .6 The fine-structure constant from the lepton masses

The fine-structure constant belongs in this counter-calculation because in FFGFT it is a lepton-mass statement: Doc. 011 sets  $\alpha = \xi(E_0/\text{MeV})^2$  with the structural energy scale  $E_0 = \sqrt{m_e m_\mu}$  and the declared fractal correction,  $\alpha^{-1} = (7500/E_0^2) K_{\text{frak}}$ . From the data side: the measured  $\alpha$  requires  $E_0^{\text{req}} = 7.39798$  MeV, the geometric mean gives 7.34788 MeV – ratio 1.006819, against  $K_{\text{frak}}^{-1/2} = 1.006734$ . The declared correction factor thus closes the gap up to  $8 \cdot 10^{-5}$ ; the closed relation

$$\alpha = \frac{\xi m_e m_\mu}{K_{\text{frak}} \text{MeV}^2}, \quad K_{\text{frak}} = 1 - 100 \xi, \quad (17)$$

gives  $1/\alpha = 137.059$  against measured 137.035999. This agreement is a property of the anchor  $E_0$ , which provides the connection to SI, not an independent prediction of  $\alpha$ .

The decisive point here is that  $\alpha$  does *not* run through a fitted bridge constant. Two closely related routes lead to the characteristic energy  $E_0$  (A130): Route 1 is the bare geometric mean of the measured masses,  $E_0 = \sqrt{m_e m_\mu} = 7.348$  MeV, and gives  $1/\alpha = 138.9$  – in this variant  $K_{\text{frak}}$  cancels (ratio calculation, Doc. 122); Route 2 is the same mean with the fractal correction,  $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.397$  MeV, and gives  $1/\alpha = 137.06$  ( $+0.017\%$ ).  $K_{\text{frak}} = 1 - 100 \xi$  comes from  $\xi$ , not from  $\alpha$  – no back-calculation from  $\alpha$ . Read the other way, the measured value of  $\alpha$  fixes the factor between the two routes:  $K_{\text{frak}} = \xi m_e m_\mu \alpha^{-1} = 0.98650$ , against  $1 - 100 \xi = 0.98667$  to  $1.7 \cdot 10^{-4}$ . Comparing the two routes with  $\alpha$  thus confirms the value  $1 - 100 \xi$  up to this residual; it is not an  $\alpha$ -free measurement of  $K_{\text{frak}}$ . That *the same* factor that carries the damped X-flip and the  $T_{\text{CMB}}$  correction in the corpus closes the gap between Route 1 and the  $E_0$  required by  $\alpha$  up to this residual is a cross-sector consistency – not a fit.

### The $\alpha$ residual and the dressing layer

Taking the  $\alpha$  measured to  $1.5 \cdot 10^{-10}$  as exact, a residual remains between Route 2 and the  $E_0 = \sqrt{\alpha/\xi} = 7.3980$  MeV required by  $\alpha$ . The  $K_{\text{frak}}$  bridge closes 98.8% of the gap between Route 1 and the  $E_0$  required by  $\alpha$ ,  $(7.39736 - 7.34788)/(7.39798 - 7.34788) = 0.988$ ; a residual of  $8.4 \cdot 10^{-5}$  remains at the  $E_0$  level ( $1.7 \cdot 10^{-4}$  in  $K_{\text{frak}}$ ). Measured against the precision of the masses ( $1.1 \cdot 10^{-8}$  at the  $E_0$  level), this residual is  $\sim 7700\sigma$  large – too large to be mass measurement noise.

Where it sits cannot be decided from the  $\alpha$  sector alone: both routes use  $m_e$  and  $m_\mu$ , and the residual may lie in the pole masses (QED dressing) just as well as in the coefficient of  $K_{\text{frak}}$  (101.2 instead of 100). Compared with the  $\theta$  side (Section .5), where  $\theta_{\text{ref}}$  is obtained from  $m_\mu/m_e$  and the  $\sim 450\sigma$  in the  $\mu/e$  channel are half self-reference, the  $\alpha$  residual is therefore not a stronger, mode-independent witness but an equal-ranking indication of a layer between the geometrically fundamental quantity and the measured pole mass, one that leaves the attribution open. The obligation – to *compute* the dressing rather than name it as a residual – remains open.

Conceptually – almost without computational consequence, yet central to the architecture – the following holds: in the extended natural unit system  $\alpha = 1$  (Docs. 011/014, charge dimensionless); the SI value  $1/137.036$  is bookkeeping of the unit translation, not a fundamental coupling. The same architecture carries gravitation:  $G = \xi^2/(4m_{\text{char}})$  times conversion (Docs. 012/013) makes  $G$  a bookkeeping quantity relative to SI – reproducing the SI value is calibration per R40; only the forward chain is testable. The structural gain lies elsewhere: a coupling that is only bookkeeping need not be quantised – the quantum-gravity problem, on which most extensions of the Standard Model founder, does not arise in this architecture.

## .7 Two layers that do not need each other: what $2/9$ provides and what the SI conversion provides

Two questions that are easily conflated must be separated here.

*First: is  $2/9$  needed to compute the masses?* No. The historical and still valid route is the  $\xi$  ladder (Docs. 006/046):  $m_i = r_i \xi^{p_i} v$  with  $r \in \{4/3, 16/5, 25/9\}$  and  $p \in \{3/2, 1, 2/3\}$  yields the masses from  $\xi$  and the ladder numbers alone, with  $2/9$  entering nowhere. The structural number  $2/9$  appears only from Doc. 282 onward and in a different role: not as an input generating the masses, but as an *order discovered within them* – the three  $\sqrt{m_k}$  are eigenvalues of a  $Z_3$  circulant with phase angle  $2/9$ .  $2/9$  replaces the formerly fitted ladder coefficient by a pure number; it is compression and explanation, not a prerequisite. One could delete  $2/9$  from the mass sector and the ladder would keep computing – what would be lost is the *insight* into why  $Q$  is exactly  $2/3$ , not the numbers.  $2/9$  thus belongs to the explanatory layer, not the computational one.

*Second: does this counter-calculation confirm that the SI conversion is necessary and correct?* In two separate degrees. As *necessary* it confirms clearly: every absolute number ( $m_\tau$  in MeV,  $\alpha = 1/137$ ,  $G$ ) requires the bridge constant to reach SI, while the ratios ( $m_\mu/m_e$ ) do not. This two-level structure is exactly P40, and the  $450\sigma$  finding is its sharpest evidence: it shows that the measured pole-mass level (SI-defined, QED-dressed) is a *different* level from the geometric one – without the conversion step the difference would not exist. As *correct in the sense of validated to measurement precision*, however, it does *not* confirm: the conversion closes about 98.8% of the gap at  $\alpha$  (through the same factor  $K_{\text{frak}}$  that also operates in other sectors), but the remaining  $\sim 10^{-4}$  stays open. The SI conversion is therefore structurally necessary and consistent across sectors, but it is calibration, not a test passed to the decimal.

The real finding lies in the independence: the ladder computes masses without  $2/9$ , the circulant explains  $Q$  without the ladder, the SI conversion translates without  $2/9$ . That all three nonetheless share the same  $\xi$  and the same factor  $K_{\text{frak}}$  is the consistency of the framework – not a dependence of one layer on another.

## .8 Placement

The counter-calculation from the data side confirms the two-level architecture of the corpus without adding to it or sparing it anything. At the circulant level the data return the two structural numbers at the  $10^{-5}$  level; the precision analysis separates two findings: in the  $\tau$ -limited (thick) direction a single  $\tau$  tension ( $0.9\sigma$  with PDG 2022,  $0.43\sigma$  with PDG 2024), carrying the pre-booked prediction  $m_\tau = 1776.968$  MeV – and in the  $\mu/e$ -precise (thin) direction an already measured  $10^{-5}$  offset that excludes the *exactness* of the pair  $(\sqrt{2}, 2/9)$  and is booked as open fine structure. At the ladder level the data return  $\xi$  at 1–3 %, with honestly open residuals showing no recognisable  $K_{\text{frak}}$  pattern. What this calculation does *not* show: no derivation of  $\theta = 2/9$  (its dynamical locus remains the open question of the bridge work), no sharpening of the ladder, no new physics. It shows that the declared claims of both levels are independently reproducible starting from the PDG data – in the direction in which nothing can be fitted.

### The residual as the signature of incompleteness

The counter-calculation has sharpened the statements – from “two pure numbers, no fit” to measured quantities with known decimals and known limits. Yet at every level a residual remains: the  $10^{-7}$  offset in  $\theta$ , the 1–3 % of the ladder, the open  $\delta$  fine value (Doc. 291), the  $K_{\text{frak}}$  pattern that does not close across the ladder stages. The  $\alpha$  point also carries a residual: the  $K_{\text{frak}}$  required by  $\alpha$  lies  $1.7 \cdot 10^{-4}$  from  $1 - 100\xi$  (see above). None of these residuals is a *contradiction* – the theory holds everywhere to its respective precision – but none is *zero*. A residual that shrinks the closer one looks without vanishing is the signature of an approximation to something deeper, not of a completed foundation. The residual is therefore not an error but the honest evidence that the theory is *not complete*.

What is remarkable is *where* it sits. The geometric core carries no residual:  $Q = 2/3$  is purely geometric and practically exact; in the extended natural system  $\alpha = 1$ ; the ladder ratios are rational numbers. The residual arises throughout *only* at the transition into the measured: the  $450\sigma$  break only at the pole-mass level, the ladder deviation only in the bridge constant. The  $\alpha$  conversion fits in as well: there the measured value of  $\alpha$  fixes the correction depth ( $K_{\text{frak}} = 0.98650$ ), and the residual of  $1.7 \cdot 10^{-4}$  against  $1 - 100\xi$  arises only in the transition to the measured quantity. The residual thus lives not in the geometric core but in the *translation layer* – systematically, not randomly distributed. This localises the incompleteness precisely: not in  $\xi$ , not in the circulant, but in the not-yet-derived fractal or QED dressing. The theory would be complete only once it *computes* this residual instead of absorbing it into a bridge constant – once  $K_{\text{frak}}$  not merely fits but follows generation-dependently from the geometry. The measure for this is expressly *not* a point-exact landing on the last decimal: a geometric derivation is not meant to hit the QED-dressed pole mass digit by digit – that would be neither possible nor the goal – but to fix the underlying structure far enough that the predictions stay *within the tolerances* to which the quantities can be determined at all. Completeness here means: no residual left that protrudes *beyond* those tolerances and is closed only by a free parameter. Until then, each such residual is the pointer to the next open derivation, against which completeness – in the sense of tolerance, not of the last digit – will be measured.